SAHIL PATEL

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Technical Skills

Languages: Kotlin, Java, Python
Android: Android SDK, Jetpack Compose, XML, Coroutines, Flow, Hilt, Room, WorkManager, Paging 3
Architecture: MVVM, MVI, Clean Architecture, Repository Pattern, Use Cases, UDF, Modularization, SOLID
Networking & Data: REST APIs, WebSockets, Retrofit, OkHttp Interceptors, AES-256 Encryption, UDP
AI/ML & Audio: TensorFlow Lite, MediaPipe, On-Device ML, Audio Processing, Frequency Analysis, Masking, Notch Filtering
Performance: Android Profiler, CPU/Memory Profiling, Baseline Profiles, App Startup Optimization, LeakCanary, ANR Analysis

Work Experience

Senior Android Engineer

Jan 2022 – Present

WeHear Innovations Pvt. Ltd.

Ahmedabad, India

WeHearApp

- Led **Android modernization** from Java/XML → Kotlin/XML → Jetpack Compose, introducing MVVM, Clean Architecture, StateFlow, and lifecycle-aware state management to improve scalability, maintainability, and UI consistency.
- Architected the **real-time translation platform** for one-to-one, group, and live translation using Repository, Use Cases, Room, Coroutines, Flow, and TranslationCoordinator, enabling scalable session and segment-level data management.
- Optimized **translation response pipelines** using structured concurrency, cancellation, debouncing, caching, controlled parallelism, and response ordering, achieving **60% faster response time** across measured scenarios.
- **Collaborated on-site with firmware engineers in China** to co-design and iterate BLE communication protocols across 5+ firmware versions — defining 42+ command types, packet structures, acknowledgment flows, and error recovery contracts, ensuring deterministic alignment between Android application layer and embedded firmware constraints.
- Led **field validation in China**, collecting feedback on translation accuracy, latency, speech segmentation, and UX; converted real-world findings into feedback-driven improvements across translation workflows, response handling, and incremental rendering.
- Improved **translation UX based on field feedback**, refining speech segmentation, partial-result presentation, loading states, retry handling, and response synchronization to provide more predictable real-time translation experiences.
- **Invented PairHub™** — an intelligent multi-device switching system enabling seamless transition between previously connected devices without manual Bluetooth re-pairing, delivering a frictionless ecosystem-like connectivity experience — supporting up to **4 simultaneously paired wearables** and improving device switch latency by ∼**55%**.
- Engineered **process death resilience** using SavedStateHandle and persistent state serialization for seamless BLE session recovery after system-initiated process termination; implemented **DataStore (Proto)** for type-safe async device preferences; **Paging 3** with Room RemoteMediator for conversation history; and **Baseline Profiles** reducing cold start time by ∼**35%**.

NUConnect – BLE Companion Platform for Smart Hearing Aids

- Engineered a unified **Android BLE communication layer** supporting **6 hearing-aid device families** through vendor-specific GATT mappings, capability abstraction, connection-state management, characteristic operations, and notifications.
- Optimized **BLE discovery and connection workflows** using targeted scan filters, connection-priority tuning, timeout policies, and automatic reconnection, reducing **scan/connection time by 70%** and improving device onboarding reliability.
- Engineered **bidirectional BLE command orchestration** for independent left/right hearing-aid channels using prioritized read/write queues, command serialization, acknowledgments, and state synchronization to prevent collisions and stale device state.
- Built real-time **hearing-aid configuration workflows** for volume, listening modes, noise reduction, multi-device management, and profile synchronization, enabling reliable GATT payload exchange across supported devices.
- Developed a **remote configuration assistant** using patient pre-assessment questionnaires and configurable profile rules to recommend and apply personalized device settings, enabling remote fitting workflows and reducing **clinic visits by approximately 60%**.

NUScope – Android Otoscope & Medical Device Communication Platform

- Reverse-engineered the **otoscope communication protocol** by tracing command sequences, decoding device responses, and validating undocumented operations with the China hardware team for production integration.
- Engineered **real-time UDP video streaming over Wi-Fi** with packet sequencing, frame reassembly, jitter buffering, adaptive frame dropping, and stream synchronization, achieving ∼99% smooth frame delivery under unstable network conditions.
- Implemented **Wi-Fi Station-mode networking** to maintain local otoscope connectivity while coordinating internet access for cloud services without interrupting active medical-device sessions.
- Developed **network recovery and adaptive routing** for interface changes, packet loss, device reachability, and network transitions; combined with **offline-first persistence** to maintain reliable device operations during intermittent connectivity.

WeConverse – Real-Time Multilingual Communication Platform

- **Led cross-functional engineering team** (iOS, Android, Backend, Product) to deliver WeConverse as the **official communication partner for BNI Global Convention 2024, Hawaii** — streaming live translated audio to **3,000+ inter-country participants** across **100+ languages** in real time.
- Engineered WebSocket-based messaging with persistent bidirectional communication and AES-256 encryption, enabling low-latency chat and reliable message delivery during live events.
- Optimized high-volume event archives using Paging, incremental loading, request deduplication, and cache-aware data retrieval, reducing memory pressure and keeping large historical datasets responsive.
- Implemented event-driven data synchronization for live sessions, coordinating asynchronous API responses, real-time updates, and UI state transitions without blocking the main thread during active events.

Tinni-X – Personalized Tinnitus Assessment, Therapy & Audio Processing Platform

- Engineered a **tinnitus assessment workflow** around a testing algorithm to establish each user's **baseline tinnitus level**, supporting self-guided evaluation and personalized therapy recommendations across **50+ assessment sessions**.
- Designed a **personalized audio synthesis pipeline** combining curated sounds and music with configurable mixing parameters, enabling users to create custom therapy soundscapes based on assessment results and listening preferences.
- Applied **Adaptive Tinnitus Notch Filtering (ATNF)** as an optional post-processing stage, allowing users to generate therapy audio with **notching enabled or disabled** around patient-specific tinnitus frequencies.
- Created a **baseline-driven therapy scheduling engine** mapping assessment results to session duration, therapy frequency, progression, and recommended usage schedules, reducing manual therapy configuration by ∼**55%**.
- Built **therapy playback and usage analytics** to track session duration, selected sound profiles, playback history, and therapy configuration, supporting adherence monitoring and personalized progression.

Education

Government Engineering College, Modasa — Bachelor's in Information Technology — Aug 2019 – May 2022
R.C Technical Institute — Diploma in Information Technology — Aug 2016 – May 2019

Open Source & Personal Projects

detekt — Kotlin Static Analysis Tool — GitHub

Kotlin • Static Analysis • ktlint • JVM 17

- Resolved private-extension false-positive detection issues in detekt through PRs #9450 and #9453, improving analysis accuracy for Kotlin codebases.
- Enhanced JVM 17 compatibility by contributing ktlint wrapper support, addressing tooling integration requirements for modern Kotlin projects.

DuckDuckGo Android — Open Source Contribution — GitHub

Kotlin • Android SDK • Storage Access Framework

- Contributed PR #8844 to improve download-location handling through Android's Storage Access Framework, strengthening file-access behavior across supported storage providers.

ReachMe — Privacy-First QR Vehicle Communication Platform — GitHub

Kotlin • Jetpack Compose • Node.js • Express • MongoDB • Firebase • Socket.IO

- Engineered a QR-based communication platform that lets users contact vehicle owners without exposing personal phone numbers.
- Designed QR registration, unique ID generation, authentication, vehicle management, and activation workflows across Android and REST APIs.
- Enabled real-time text, voice, push notifications, and one-way low-latency audio using Firebase and Socket.IO for asynchronous and live communication.

Styledrobe — Offline AI Wardrobe Recommendation System — Google Play

Kotlin • TensorFlow Lite • Jetpack Compose • Room • Hilt

- Developed an on-device garment classification and outfit recommendation pipeline with TensorFlow Lite, removing cloud dependency from ML inference.
- Optimized image preprocessing and local inference workflows to deliver privacy-preserving recommendations without uploading wardrobe data.
- Structured the application with Clean Architecture, Hilt, Room, and Jetpack Compose; published the production app on Google Play.

Zbukk — AI-Powered Android Accessibility Platform — Google Play

Kotlin • CameraX • ML Kit • Android Accessibility Service

- Developed hands-free Android navigation using real-time eye-blink detection and computer vision to enable touch-free device interaction.
- Connected CameraX and ML Kit with Android Accessibility Service to translate detected gestures into scrolling actions across third-party applications.
- Streamlined on-device vision processing for responsive gesture detection while keeping camera data local to the device.